A CLOSED FORM AND A LUCAS CONGRUENCE FOR APÉRY'S COMPANION SEQUENCE

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Abstract. We give a two-term closed form for the numerators b_n of Apéry's approximations to $\zeta(3)$, a Lucas-type congruence $p^3 b_{ap+r} \equiv b_a\, a_r \pmod p$ for them, and a correction to a published integrality claim. Every theorem and lemma below is proved in Lean 4 and checked by its kernel, and is followed by the name of the declaration that proves it; the one conjecture is labelled as such.

1. Introduction

Apéry's proof that $\zeta(3) \notin \mathbb{Q}$ rests on the pair of sequences $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and a companion b_n with $b_n/a_n \to \zeta(3)$. The first sequence is arithmetically well understood: it satisfies a Lucas congruence $a_{ap+r} \equiv a_a a_r \pmod p$ (Gessel [2]), so it respects base-p digits. The companion b_n has attracted no such statement, and it is usually written with a nested alternating inner sum, which makes it awkward to handle modulo p.

We report three things.

First, a closed form (Theorem 2.6): b_n is the same binomial sum as a_n weighted by a *two-term* harmonic expression,

$$b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \big(2H_n^{(3)} - H_k^{(3)}\big), \qquad H_m^{(r)} = \sum_{j=1}^{m} j^{-r}.$$

The nested sum disappears.

Second, a Lucas-type congruence for the companion itself (Theorem 2.7):

$$p^3\, b_{ap+r} \equiv b_a\, a_r \pmod p \qquad (a, r < p).$$

The factor p^3 is the weight of $\zeta(3)$. The same argument gives the analogous congruence for Franel's companion at weight 2 (Theorem 2.8).

Third, a correction. Brown and Zudilin [1] record $d_n^5 P_n \in \mathbb{Z}$ for the numerators of their cellular approximations to $\zeta(5)$, where $d_n = \operatorname{lcm}(1, \dots, n)$. Their own printed value $P_2 = 1190161/384$ does not satisfy this. At $n = 2$ the defect is exactly $12 = 2^2 \cdot 3$ (Theorem 3.2); consequently any constant c for which $c\, d_n^5 P_n$ is always integral is divisible by 12. Whether 12 suffices for every n we state only as Conjecture 3.3: it is supported by computation and by a proof for the primes $p \geq 5$ given elsewhere, and it is not formalised.

A second, benign slip in the same author's earlier work is worth recording, though we do not formalise it: equation (8.7) of [5], defining the rational function $R(t)$, omits the factor $(h_0 + 2t)$ present in (8.2). That factor is the only one antisymmetric under $t \mapsto -t - h_0$, and it is what forces the coefficients of the even zeta values to vanish; without it the parity structure of the linear forms is reversed.

Throughout, p is prime, and for $x, y \in \mathbb{Q}$ we write $x \equiv y \pmod p$ to mean $v_p(x - y) \geq 1$. `PInt, PDvd, PCong`

2. Apéry's companion

Definition 2.1. $A(n,k) = \dbinom{n}{k}^2 \dbinom{n+k}{k}^2$ and $a_n = \sum_{k=0}^{n} A(n,k)$. `A, apery`

Lemma 2.2. *For $r, s < p$, $A(ap + r,\ bp + s) \equiv \binom{a}{b}^2 \binom{a+b}{b}^2 A(r, s) \pmod p$.* *A_digits*

This is Lucas' theorem applied to each binomial, together with the fact that a carry $r + s \geq p$ makes both sides vanish. `choose_digits, choose_carry_zero`

Definition 2.3. $P(n) = 34n^3 + 51n^2 + 27n + 5$, and b_n is defined by

$$b_0 = 0, \qquad b_1 = 6, \qquad (n + 2)^3 b_{n+2} = P(n + 1)\, b_{n+1} - (n + 1)^3 b_n.$$

`Ppol, bApery`

This is Apéry's recurrence; $b_2 = 351/4$ and $b_3 = 62531/36$.

Definition 2.4. $\tilde{b}_n = \sum_{k=0}^{n} A(n, k)\big(2H_n^{(3)} - H_k^{(3)}\big)$. `Harm, bMin`

Lemma 2.5. *$(m + 2)^3 \tilde{b}_{m+2} - P(m + 1)\, \tilde{b}_{m+1} + (m + 1)^3 \tilde{b}_m = 0$ for all $m \geq 0$.* *bMin_rec*

The proof is a single telescoping step: applying the recurrence operator to the summand of $\tilde{b}$ produces an exact difference $\Psi(m, k + 1) - \Psi(m, k)$, whose sum collapses to the boundary values $\Psi(m, 0) = \Psi(m, m+3) = 0$. The two Zeilberger-type certificates involved are polynomial identities in $\mathbb{Q}[m, k]$. `star, star_pre`

Theorem 2.6. *$\tilde{b}_n = b_n$ for all n; that is,*

$$\sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \big(2H_n^{(3)} - H_k^{(3)}\big) = b_n.$$

bMin_eq_bApery, apery_b_harmonic_closed_form

Theorem 2.7. *For $a, r < p$, $p^3\, b_{ap+r} \equiv b_a\, a_r \pmod p$.* *bApery_lucas*

The weight 3 enters as follows: modulo p the summands with $k \not\equiv 0$ contribute nothing after multiplication by p^3, while the layer $k = jp$ reproduces $H^{(3)}$ at $\lfloor \cdot /p \rfloor$, since $p^3 \sum_{m \leq y,\ p|m} m^{-3} = \sum_{j \leq y/p} j^{-3}$. The remaining digit factorisation is Lemma 2.2.

Theorem 2.8. *Let $f_n = \sum_k \binom{n}{k}^3$ and $\beta_n = \sum_k \binom{n}{k}^3 \big(\frac{1}{4}H_k^{(2)} + \frac{3}{4}(H_k^{(1)})^2 - \frac{3}{4}H_k^{(1)} H_{n-k}^{(1)}\big)$. Then for $p \neq 2$ and $a, r < p$,*

$$p^2\, \beta_{ap+r} \equiv \beta_a\, f_r \pmod p.$$

franel, bFranel, bFranel_lucas

3. A denominator defect

Definition 3.1. $d_n = \operatorname{lcm}(1, \ldots, n)$, and $P_2 = 1190161/384$ is the value printed in [1]. `dlcm, BZ_P2`

Theorem 3.2. *$d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$, while $12\, d_2^5 P_2 = 1190161 \in \mathbb{Z}$; and if $c \in \mathbb{N}$ satisfies $c\, d_2^5 P_2 \in \mathbb{Z}$, then $12 \mid c$.* *BZ_P2_not_integral, BZ_P2_factor_twelve, BZ_twelve_minimal*

Here $d_2 = 2$ and $1190161 \equiv 1 \pmod{12}$, so at $n = 2$ the defect is exactly $2^2 \cdot 3$. Specialising to $n = 2$, any $c \in \mathbb{N}$ with $c\, d_n^5 P_n \in \mathbb{Z}$ for all n is divisible by 12.

Conjecture 3.3. $12\, d_n^5 P_n \in \mathbb{Z}$ for every $n \geq 1$.

Theorem 3.2 shows that no smaller constant can serve. Conjecture 3.3 itself is *not* formalised: it is supported by exact computation, and its restriction to the primes $p \geq 5$ is proved elsewhere, conditionally on two finite identities verified but not machine-checked.

4. The development

Lean 4 (`v4.33.0-rc1`) with Mathlib `cd580e54`; ten modules, 2107 lines; no `sorry` and no `native_decide`. Besides the results above it contains Gessel's congruence $a_{ap+r} \equiv a_a a_r$ [2] and its analogue for the Brown–Zudilin double sum

$$Q_n = \sum_{k,l} \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n},$$

in both cases with the base-p digit-product form (`apery_lucas`, `Q_lucas`). Every declaration cited above satisfies

$$\texttt{\#print axioms} \longrightarrow [\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}].$$

Build with `lake exe cache get && lake build`.

References

[1] F. Brown and W. Zudilin, *On cellular rational approximations to* $\zeta(5)$, preprint `arXiv:2210.03391v3` (2022).

[2] I. M. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), no. 3, 362–368.

[3] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381. `arXiv:1910.09336`.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28, LNCS 12699, 625–635.

[5] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.